



Subject Outline | Stage 1



SUBJECT OUTLINE CHANGES

Below are the current changes to the subject outline. Teachers are encouraged to **explore the changes in detail** and make relevant adjustments to their teaching, learning, and assessment programs.

From 2024	To 2025 onwards	page
Stage 1		
Evidence of Learning – 10-credit subject Four assessments	Evidence of Learning – 10-credit subject Three or four assessments	53
Evidence of Learning – 20-credit subject Eight assessments • two investigations with a focus on science as a human endeavour	Evidence of Learning – 20-credit subject Six or seven assessments • at least one investigation with a focus on science as a human endeavour	53 , 54
Evidence of Learning – 20-credit subject • at least two skills and applications tasks under direct teacher supervision	Evidence of Learning – 20-credit subject • at least one skills and applications task under direct teacher supervision	57
Assessment Type 1: Investigations Folio • Evidence of deconstruction . . . to a maximum of four sides of an A4 page.	Assessment Type 1: Investigations Folio • Evidence of deconstruction . . . to a maximum of four sides of an A4 page (minimum font size 10).	55
Performance standards – Investigation, Analysis and Evaluation (IAE1) – A to C grade bands.	Inclusion of 'justification' to the IAE1 A to C grade bands: • A grade band – Critically deconstructs a problem and designs a logical and coherent physics investigation with detailed justification . • B grade band – Logically deconstructs a problem and designs a well-considered and clear physics investigation with reasonable justification . • C grade band – Deconstructs a problem and designs a considered and generally clear physics investigation with some justification .	59

Published by the SACE Board of South Australia,
11 Waymouth Street, Adelaide, South Australia 5000

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First published 2016

Published online October 2016

Reissued for 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

ISBN 978 1 74102 816 4 (online Microsoft Word version)

ref: A1441152

This subject outline is accredited for teaching at Stage 1 from 2017

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INTRODUCTION

SUBJECT DESCRIPTION

Physics is a 10-credit subject or a 20-credit subject at Stage 1 and a 20-credit subject at Stage 2.

The study of Physics is constructed around using qualitative and quantitative models, laws, and theories to better understand matter, forces, energy, and the interaction among them. Physics seeks to explain natural phenomena, from the subatomic world to the macrocosmos, and to make predictions about them. The models, laws, and theories in physics are based on evidence obtained from observations, measurements, and active experimentation over thousands of years.

By studying physics, students understand how new evidence can lead to the refinement of existing models and theories and to the development of different, more complex ideas, technologies, and innovations.

Through further developing skills in gathering, analysing, and interpreting primary and secondary data to investigate a range of phenomena and technologies, students increase their understanding of physics concepts and the impact that physics has on many aspects of contemporary life.

By exploring science as a human endeavour, students develop and apply their understanding of the complex ways in which science interacts with society and investigate the dynamic nature of physics. They explore how physicists develop new understanding and insights and produce innovative solutions to everyday and complex problems and challenges in local, national, and global contexts.

In Physics, students integrate and apply a range of understanding, inquiry, and scientific thinking skills that encourage and inspire them to contribute their own solutions to current and future problems and challenges. Students also pursue scientific pathways, for example, in engineering, renewable energy generation, communications, materials innovation, transport and vehicle safety, medical science, scientific research, and the exploration of the universe.

CAPABILITIES

The capabilities connect student learning within and across subjects in a range of contexts. They include essential knowledge and skills that enable people to act in effective and successful ways.

The SACE identifies seven capabilities. They are:

- literacy
- numeracy
- information and communication technology (ICT) capability
- critical and creative thinking
- personal and social capability
- ethical understanding
- intercultural understanding.

Literacy

In this subject students extend and apply their literacy capability by, for example:

- interpreting the work of scientists across disciplines using physics knowledge
- critically analysing and evaluating primary and secondary data
- extracting physics information presented in a variety of modes
- using a range of communication formats to express ideas logically and fluently, incorporating the terminology and conventions of physics
- synthesising evidence-based arguments
- communicating appropriately for specific purposes and audiences.

Numeracy

In this subject students extend and apply their numeracy capability by, for example:

- solving problems using calculations and critical thinking skills
- measuring with appropriate instruments
- recording, collating, representing, and analysing primary data
- accessing and interpreting secondary data
- identifying and interpreting trends and relationships
- calculating and predicting values by manipulating data and using appropriate scientific conventions.

Information and communication technology (ICT) capability

In this subject students extend and apply their ICT capability by, for example:

- locating and accessing information
- collecting, analysing, and representing data electronically
- modelling concepts and relationships
- using technologies to create new ways of thinking about science
- communicating physics ideas, processes, and information

- understanding the impact of ICT on the development of physics and its application in society
- evaluating the application of ICT to advance understanding and investigations in physics.

Critical and creative thinking

In this subject students extend and apply critical and creative thinking by, for example:

- analysing and interpreting problems from different perspectives
- deconstructing a problem to determine the most appropriate method for investigation
- constructing, reviewing, and revising hypotheses to design investigations
- interpreting and evaluating data and procedures to develop logical conclusions
- analysing interpretations and claims, for validity and reliability
- devising imaginative solutions and making reasonable predictions
- envisaging consequences and speculating on possible outcomes
- recognising the significance of creative thinking on the development of physics knowledge and applications.

Personal and social capability

In this subject students extend and apply their personal and social capability by, for example:

- understanding the importance of physics knowledge on health and well-being, both personally and globally
- making decisions and taking initiative while working independently and collaboratively
- planning effectively, managing time, following procedures effectively, and working safely
- sharing and discussing ideas about physics issues developments, and innovations while respecting the perspectives of others
- recognising the role of their own beliefs and attitudes in gauging the impact of physics in society
- seeking, valuing, and acting on feedback.

Ethical understanding

In this subject students extend and apply their ethical understanding by, for example:

- considering the implications of their investigations on organisms and the environment
- making ethical decisions based on an understanding of physics principles
- using data and reporting the outcomes of investigations accurately and fairly
- acknowledging the need to plan for the future and to protect and sustain the biosphere
- recognising the importance of their responsible participation in social, political, economic, and legal decision-making.

Intercultural understanding

In this subject students extend and apply their intercultural understanding by, for example:

- recognising that science is a global endeavour with significant contributions from diverse cultures
- respecting and engaging with different cultural views and customs and exploring their interaction with scientific research and practices
- being open-minded and receptive to change in the light of scientific thinking based on new information
- understanding that the progress of physics influences and is influenced by cultural factors.

ABORIGINAL AND TORRES STRAIT ISLANDER KNOWLEDGE, CULTURES, AND PERSPECTIVES

In partnership with Aboriginal and Torres Strait Islander communities, and schools and school sectors, the SACE Board of South Australia supports the development of high-quality learning and assessment design that respects the diverse knowledge, cultures, and perspectives of Indigenous Australians.

The SACE Board encourages teachers to include Aboriginal and Torres Strait Islander knowledge and perspectives in the design, delivery, and assessment of teaching and learning programs by:

- providing opportunities in SACE subjects for students to learn about Aboriginal and Torres Strait Islander histories, cultures, and contemporary experiences
- recognising and respecting the significant contribution of Aboriginal and Torres Strait Islander peoples to Australian society
- drawing students' attention to the value of Aboriginal and Torres Strait Islander knowledge and perspectives from the past and the present
- promoting the use of culturally appropriate protocols when engaging with and learning from Aboriginal and Torres Strait Islander peoples and communities.

HEALTH AND SAFETY

It is the responsibility of the school to ensure that duty of care is exercised in relation to the health and safety of all students and that school practices meet the requirements of the *Work Health and Safety Act 2012*, in addition to relevant state, territory, or national health and safety guidelines. Information about these procedures is available from the school sectors.

The following safety practices must be observed in all laboratory work:

- Use equipment only under the direction and supervision of a teacher or other qualified person.
- Follow safety procedures when preparing or manipulating apparatus.
- Use appropriate safety gear when preparing or manipulating apparatus.

Particular care must be taken when using electrical apparatus, ionising and non-ionising radiation, and lasers, but care must not be limited to these items.

LEARNING SCOPE AND REQUIREMENTS

LEARNING REQUIREMENTS

The learning requirements summarise the knowledge, skills, and understanding that students are expected to develop and demonstrate through their learning in Stage 1 Physics.

In this subject, students are expected to:

1. apply science inquiry skills to deconstruct a problem and design and conduct physics investigations, using appropriate procedures and safe, ethical working practices
2. obtain, record, represent, analyse, and interpret the results of physics investigations
3. evaluate procedures and results, and analyse evidence to formulate and justify conclusions
4. develop and apply knowledge and understanding of physics concepts in new and familiar contexts
5. explore and understand science as a human endeavour
6. communicate knowledge and understanding of physics concepts, using appropriate terms, conventions, and representations.

CONTENT

Physics is a 10-credit or a 20-credit subject at Stage 1.

The topics in Stage 1 Physics provide the framework for developing integrated programs of learning through which students extend their skills, knowledge, and understanding of the three strands of science.

The three strands of science to be integrated throughout student learning are:

- science inquiry skills
- science as a human endeavour
- science understanding.

The topics for Stage 1 Physics are:

- Topic 1: Linear motion and forces
- Topic 2: Electric circuits
- Topic 3: Heat
- Topic 4: Energy and momentum
- Topic 5: Waves
- Topic 6: Nuclear models and radioactivity.

For a 10-credit subject, students study a selection of concepts from at least three of these topics.

For a 20-credit subject, students study a selection of concepts from all six topics.

The topics selected can be sequenced and structured to suit individual groups of students. Topics can be studied in their entirety or in part, taking into account student interests and preparation for pathways into future study of physics. For example, Topic 2: Electric circuits is valuable for students intending to seek an electrical apprenticeship and Topic 3: Heat can prepare students for a VET course in refrigeration.

Note that the topics are not necessarily designed to be of equivalent length — it is anticipated that teachers may allocate more time to some than others.

In designing a Stage 1 Physics program for students who intend to study Physics at Stage 2, the information in the following table should be taken into account. This table shows Stage 1 subtopics that introduce key ideas that are later used in the specified Stage 2 subtopics.

Stage 1	Stage 2
1.1 Motion under constant acceleration	1.1 Projectile motion
	1.2 Forces and momentum
	1.3 Circular motion and gravitation
	2.2 Motion of charged particles in electric fields
1.2 Forces	1.2 Forces and momentum
	1.3 Circular motion and gravitation
	2.1 Electric fields
	2.2 Motion of charged particles in electric fields
	2.4 Motion of charged particles in magnetic fields
2.1 Potential difference and electric current	2.1 Electric fields
	2.3 Magnetic fields
	2.5 Electromagnetic induction
4.1 Energy	2.2 Motion of charged particles in electric fields
	3.2 Wave–particle duality
4.2 Momentum	1.2 Forces and momentum
	3.2 Wave–particle duality
5.1 Wave model	3.1 Wave behaviour of light
	3.2 Wave–particle duality
	3.3 Structure of the atom
5.3 Light	3.1 Wave behaviour of light
	3.2 Wave–particle duality
6.1 The nucleus	3.4 Standard Model
6.2 Radioactive decay	2.4 Motion of charged particles in magnetic fields
	3.4 Standard Model

The following pages describe in more detail:

- science inquiry skills
- science as a human endeavour
- the topics for science understanding.

The descriptions of the science inquiry skills and the topics are structured in two columns: the left-hand column sets out the science inquiry skills or science understanding and the right-hand column sets out possible contexts.

Together with science as a human endeavour, the science inquiry skills and science understanding form the basis of teaching, learning, and assessment in this subject.

The possible contexts are suggestions for potential inquiry approaches and are neither comprehensive nor exclusive. Teachers may select from these and are encouraged to consider other approaches according to local needs and interests.

Within the topic descriptions, the following symbols are used in the possible contexts to show how a strand of science can be integrated:



indicates a possible teaching and learning strategy for science understanding



indicates a possible science inquiry activity



indicates a possible focus on science as a human endeavour.

Science Inquiry Skills

In Physics, investigation is an integral part of the learning and understanding of concepts, by using scientific methods to test ideas and develop new knowledge.

Practical investigations must involve a range of both individual and collaborative activities, during which students extend the science inquiry skills described in the table that follows.

Practical activities may take a range of forms, such as developing or using models and simulations that enable students to develop a better understanding of particular concepts. The activities include laboratory and field studies during which students develop investigable questions and/or testable hypotheses, and select and use equipment appropriately to collect data. The data may be observations, measurements, or other information obtained during the investigation. Students represent and analyse the data they have collected; evaluate procedures, and describe the limitations of the data and procedures; consider explanations for their observations; and present and justify conclusions appropriate to the initial question or hypothesis.

For a 10-credit subject, it is recommended that a minimum of 8–10 hours of class time involves practical activities.

For a 20-credit subject, it is recommended that a minimum of 16–20 hours of class time involves practical activities.

Science inquiry skills are fundamental to students investigating the social, ethical, and environmental impacts and influences of the development of scientific understanding and the applications, possibilities, and limitations of science. These skills enable students to critically analyse the evidence they obtain so that they can present and justify a conclusion.

Science Inquiry Skills	Possible contexts
Scientific methods enable systematic investigation to obtain measurable evidence. - Deconstruct a problem to determine and justify the most appropriate method for investigation. - Design investigations, including: - hypothesis or inquiry question - types of variables - dependent - independent - factors held constant (how and why they are controlled) - factors that may not be able to be controlled (and why not) - materials required - the method to be followed - the type and amount of data to be collected - identification of ethical and safety considerations.	Develop inquiry skills by, for example: - designing investigations that require investigable questions and imaginative solutions (with or without implementation) - critiquing proposed investigations - using the conclusion of one investigation to propose subsequent experiments - changing an independent variable in a given procedure and adapting the method - researching, developing, and trialling a method - improving an existing procedure - identifying options for measuring the dependent variable - researching hazards related to the use and disposal of physics materials - developing safety audits - identifying relevant ethical and/or legal considerations in different contexts.

Science Inquiry Skills	Possible contexts
Obtaining meaningful data depends on conducting investigations using appropriate procedures and safe, ethical working practices. - Conduct investigations, including: - selection and safe use of appropriate materials, apparatus, and equipment - collection of appropriate primary or secondary data (numerical, visual, descriptive) - individual and collaborative work.	Develop inquiry skills by, for example: - identifying equipment, materials, or instruments fit for purpose - practising techniques and safe use of apparatus - comparing resolution of different measuring tools - distinguishing between, and using, primary and secondary data.
Results of investigations are represented in a well-organised way to allow them to be interpreted. - Represent results of investigations in appropriate ways, including: - use of appropriate SI units, symbols - construction of appropriately labelled tables - drawing of graphs, including lines or curves of best fit as appropriate - use of significant figures.	Develop inquiry skills by, for example: - practising constructing tables to tabulate data, including column and row labels with units - identifying the appropriate representations to graph different data sets - selecting axes and scales, and graphing data - clarifying understanding of significant figures using, for example: http://www.astro.yale.edu/astro120/SigFig.pdf - comparing data from different sources to describe as quantitative or qualitative.
Scientific information can be presented using different types of symbols and representations. - Select, use, and interpret appropriate representations, including: - mathematical relationships, including direct or inverse proportion and exponential relationships - diagrams and multi-image representations - formulae to explain concepts, solve problems, and make predictions.	Develop inquiry skills by, for example: - writing formulae - using formulae; deriving and rearranging formulae - constructing vector diagrams - drawing and labelling diagrams - sketching field diagrams - recording images - constructing flow diagrams.

Science Inquiry Skills	Possible contexts
Analysis of the results of investigations allows them to be interpreted in a meaningful way. - Analyse data, including: - multi-image representations - identification and discussion of trends, patterns, and relationships - interpolation or extrapolation where appropriate.	Develop inquiry skills by, for example: - analysing data sets to identify trends and patterns - determining relationships between independent and dependent variables, including mathematical relationships, e.g. slope, linear, inverse relationships, where relevant. - discussing inverse and direct proportionality - using graphs from different sources (e.g. CSIRO or the Australian Bureau of Statistics (ABS)) to predict values other than plotted points - calculating means, standard deviations, percent error, where appropriate.
Critical evaluation of procedures and data can determine the meaningfulness of the results. - Identify sources of uncertainty, including: - random and systematic errors - uncontrolled factors. - Evaluate reliability, accuracy, and validity of results, by discussing factors including: - sample size - precision - resolution of equipment - random error - systematic error - factors that cannot be controlled.	Develop inquiry skills by, for example: - discussing how the repeating of an investigation with different materials/equipment may detect a systematic error - using an example of an investigation report to develop report-writing skills. Useful website: http://www.nuffieldfoundation.org/practical-physics/designing-and-evaluating-experiments
Conclusions can be formulated that relate to the hypothesis or inquiry question. - Select and use evidence and scientific understanding to make and justify conclusions. - Recognise the limitations of conclusions. - Recognise that the results of some investigations may not lead to definitive conclusions.	Develop inquiry skills by, for example: - evaluating procedures and data sets provided by the teacher to determine and hence comment on the limitations of possible conclusions - using data sets to discuss the limitations of the data in relation to the range of possible conclusions that could be made.

Science Inquiry Skills	Possible contexts
Effective scientific communication is clear and concise. • Communicate to specific audiences and for specific purposes using: ♦ appropriate language ♦ terminology ♦ conventions.	Develop inquiry skills by, for example: • reviewing scientific articles or presentations to recognise conventions • developing skills in referencing and/or footnoting • distinguishing between reference lists and bibliographies • practising scientific communication in written, oral, and multimedia formats (e.g. presenting a podcast or writing a blog).



Science as a Human Endeavour

The science as a human endeavour strand highlights science as a way of knowing and doing, and explores the purpose, use, and influence of science in society.

By exploring science as a human endeavour, students develop and apply their understanding of the complex ways in which science interacts with society, and investigate the dynamic nature of physics. They explore how physicists develop new understanding and insights, and produce innovative solutions to everyday and complex problems and challenges in local, national, and global contexts. In this way, students are encouraged to think scientifically and make connections between the work of others and their own learning. This enables them to explore their own solutions to current and future problems and challenges.

Students understand that the development of science concepts, models, and theories is a dynamic process that involves analysis of evidence and sometimes produces ambiguity and uncertainty. They consider how and why science concepts, models, and theories are continually reviewed and reassessed as new evidence is obtained and as emerging technologies enable new avenues of investigation. They understand that scientific advancement involves a diverse range of individual scientists and teams of scientists working within an increasingly global community of practice.

Students explore how scientific progress and discoveries are influenced and shaped by a wide range of social, economic, ethical, and cultural factors. They investigate ways in which the application of science may provide great benefits to individuals, the community, and the environment, but may also pose risks and have unexpected outcomes. They understand how decision-making about socio-scientific issues often involves consideration of multiple lines of evidence and a range of needs and values. As critical thinkers, they appreciate science as an ever-evolving body of knowledge that frequently informs public debate, but is not always able to provide definitive answers.

The key concepts of science as a human endeavour underpin the contexts, approaches, and activities in this subject, and must be integrated into all teaching and learning programs.

The key concepts of science as a human endeavour, with elaborations that are neither comprehensive nor exclusive, in the study of Physics are:

Communication and Collaboration

- Science is a global enterprise that relies on clear communication, international conventions, and review and verification of results.
- Collaboration between scientists, governments, and other agencies is often required in scientific research and enterprise.

Development

- Development of complex scientific models and/or theories often requires a wide range of evidence from many sources and across disciplines.
- New technologies improve the efficiency of scientific procedures and data collection and analysis. This can reveal new evidence that may modify or replace models, theories, and processes.

Influence

- Advances in scientific understanding in one field can influence and be influenced by other areas of science, technology, engineering, and mathematics.
- The acceptance and use of scientific knowledge can be influenced by social, economic, cultural, and ethical considerations.

Application and Limitation

- Scientific knowledge, understanding, and inquiry can enable scientists to develop solutions, make discoveries, design action for sustainability, evaluate economic, social, cultural, and environmental impacts, offer valid explanations, and make reliable predictions.
- The use of scientific knowledge may have beneficial or unexpected consequences; this requires monitoring, assessment, and evaluation of risk, and provides opportunities for innovation.
- Science informs public debate and is in turn influenced by public debate; at times, there may be complex, unanticipated variables or insufficient data that may limit possible conclusions.

Topic 1: Linear motion and forces

The study of Physics is the pursuit of understanding the physical world and the laws that govern it. One starting point for developing an understanding of matter, energy, forces, and the relationship that each has to another is the motion of physical bodies. While the motion of large objects is easily seen, the laws governing this motion can be subtle and often counterintuitive. In their study of Stage 1 Physics, students build on aspects of physics studied previously and then explore fundamental concepts and relationships in motion such as displacement, velocity, and acceleration, and the principles on which each is founded.

In the first part of this topic, students acquire the skills and understanding to describe and explain motion in a variety of formats, including algebraic and graphical representations. They use the equations of motion and various graphical methods to elicit quantitative and qualitative information about moving objects that undergo constant acceleration and hence further build their literacy and numeracy skills.

Following the study of motion under constant acceleration, students consolidate their understanding of forces and the effect that forces have on the motion of objects, using Newton's Laws of Motion.

Throughout this topic, the importance of the concepts and laws in explaining physical phenomena is emphasised and their role in providing a foundation for contemporary applications is also highlighted. Students explore the limitations of the models and ways in which concepts can inform and explain existing, developing, and emerging technologies.

Critical thinking and an understanding of linear motion and forces enable students to devise solutions and make reasonable predictions.

Subtopic 1.1: Motion under constant acceleration

In this subtopic students become familiar with examples of motion under constant acceleration and with the use of notation, units, prefixes, and representations in physics. Students develop an awareness of the differences between vertical and horizontal motion under constant acceleration, with an emphasis on motion under the acceleration due to gravity.

Science Understanding	Possible contexts	
Linear motion with constant velocity is described in terms of relationships between measurable scalar and vector quantities, including displacement, distance, speed, and velocity. - Solve problems using $v = \frac{s}{t}$. - Interpret solutions to problems in a variety of contexts. - Explain and solve problems involving the instantaneous velocity of an object. Acceleration is a change in motion. Uniformly accelerated motion is described in terms of relationships between measurable scalar and vector quantities, including displacement, speed, velocity, and acceleration. - Solve problems using equations for constant acceleration and $a = \frac{\Delta v}{\Delta t}$. - Interpret solutions to problems in a variety of contexts. - Make reasonable and appropriate estimations of physical quantities in a variety of contexts.	<i>This connects to the concept of acceleration used in Stage 1 Subtopic 1.2: Forces and Stage 2 subtopics 1.1: Projectile motion, 1.2: Forces and momentum, 1.3: Uniform circular motion and gravitation, and 2.2: Motion of charged particles in electric fields.</i> Compare current definitions of units with other systems of measurement such as imperial measure, and consider the benefits and limitations of each. Discuss the nature and difference between scalar and vector quantities, and how each explains different aspects of motion. Discuss velocity vectors for an object moving in a curved path, to understand instantaneous velocity. Investigate the physical interpretation of negative velocities and accelerations in context. Use SI units and common prefixes, such as 'kilo' (k), 'milli' (m) and 'micro' (μ), in practical activities to develop an awareness of reasonable estimates of physical quantities. Extend understanding and use of the equations $v = \frac{s}{t}$ and $a = \frac{\Delta v}{\Delta t}$. This could include the vector equations $\vec{v} = \frac{\Delta \vec{s}}{\Delta t}$ and $\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$.	
	Explore and clarify the relationship between velocity and acceleration using the computer interactive 'The Maze Game', https://phet.colorado.edu/.	

Science Understanding	Possible contexts	
	Investigate the development and use of the SI units. Analyse the significance of the development of internationally agreed definitions of absolute measures of time, mass, and distance, and the challenges facing scientists since the introduction of SI units.	
Graphical representations can be used qualitatively and quantitatively to describe and predict aspects of linear motion. - Use graphical methods to represent linear motion, including the construction of graphs showing: - position vs time - velocity vs time - acceleration vs time. - Use graphical representations to determine quantities such as position, displacement, distance, velocity, and acceleration. - Use graphical techniques to calculate the instantaneous velocity and instantaneous acceleration of an object.	Demonstrate how the gradient of a displacement vs time graph can be shown to be equivalent to the velocity of the object. Relate the gradient of a velocity vs time graph to the acceleration of the object. Use the area under the graph to determine the distance and displacement of an object. Construct different graphical representations using sections from popular movies or television shows. Those with chase scenes may be particularly effective. Graphical representations can be constructed using data from professional athletes. Construct position vs time graphs and velocity vs time graphs using trolleys on an inclined plane. Consider the accelerations of different masses. Use motion sensors or other multi-image technology to collect data. Refer to computer interactive 'The Moving Man', https://phet.colorado.edu/. Connect and investigate different graphical representations by calculating gradients and areas. For example: - use a position vs time graph to construct a velocity vs time graph - use a velocity vs time graph to construct an acceleration vs time graph	
	Work out how to determine the instantaneous velocity and instantaneous acceleration from non-linear graphs, using mathematical techniques.	

Science Understanding	Possible contexts	
Equations of motion quantitatively describe and predict aspects of linear motion. - Solve and interpret problems using the equations of motion: $v = v_0 + at$ $s = v_0 t + \frac{1}{2} at^2$ $v^2 = v_0^2 + 2as.$ Vertical motion is analysed by assuming that the acceleration due to gravity is constant near Earth's surface. The constant acceleration due to gravity near the surface of the Earth is approximately $g = 9.80 \text{ m s}^{-2}$. - Solve problems for objects undergoing vertical motion because of the acceleration due to gravity in the absence of air resistance. - Explain the concept of free-falling objects and the conditions under which free-falling motion may be approximated. - Describe qualitatively the effects that air resistance has on vertical motion. Use equations of motion and graphical representations to determine the acceleration due to gravity.	Show how equations can be derived using different methods: $v = v_0 + at$, using the definition of acceleration $s = v_0 t + \frac{1}{2} at^2$, using a velocity vs time graph $v^2 = v_0^2 + 2as$, algebraically. Calculate and analyse the acceleration due to gravity on the Moon using NASA footage showing a hammer and feather being dropped simultaneously. Use and discuss appropriate estimations. Use stop-motion animation to illustrate an understanding of motion. Footage of objects in motion may be analysed using tracking software.	
	Further develop scientific inquiry skills by investigating different aspects of projectile motion in sport. Experimentally determine the acceleration due to gravity by recording an object falling against an appropriate scale using a ticker-timer, motion sensor, or other multi-image applications. Use data to construct a velocity vs time graph. Design investigations to determine if mass has any effect on vertical acceleration. Investigate sideshow rides to measure and calculate physical quantities. Investigate what quantities make a ride enjoyable and how these are maximised. Consider factors such as g-force, acceleration, average speeds. Mobile devices could have suitable sensors and applications to record measurements.	
	Explore the principles behind different methods used to determine the speed of an object and evaluate the benefits and limitations of each method. Examples include radars and laser guns, point-to-point cameras. Decide the best location for point-to-point cameras to identify speeding vehicles.	

Science Understanding	Possible contexts	
	Investigate the methods used to determine the gait and speed of dinosaurs based on their tracks (such as that devised by Robert Alexander), which contribute to better understanding of early life on Earth. Conduct experiments and analyse data based on this work.	

Subtopic 1.2: Forces

Students apply Newton's Laws of Motion to a variety of contexts. Students investigate how these laws have influenced design and safety in different contexts such as cars, boats, submarines, playground equipment, and air and space transport. Through experiments and activities, students build a sound understanding of forces in the physical world, including those relating to various kinds of resistance and friction.

Science Understanding	Possible contexts
A force, $\vec{F}$, is any action which causes motion to change, $\vec{a}$. Uniform motion is a state of motion in which the body travels with a constant speed (in a straight line). Rest is a state of uniform motion in which the speed of the body is zero. To change the state of motion of an object, a net force must be applied.	<i>This connects to the concepts of force used in Stage 1 subtopics 4.1: Energy and 4.2: Momentum, and Stage 2 subtopics 1.2: Forces and momentum, 1.3: Circular motion and gravitation, 2.1: Electric fields, 2.2: Motion of charged particles in electric fields, and 2.4: Motion of charged particles in magnetic fields.</i> Review understanding of forces. 
Newton's Three Laws of Motion describe the relationship between the force or forces acting on an object, modelled as a point mass, and the motion of the object due to the application of the force or forces. Newton's First Law: An object will remain at rest, or continue in its motion, unless acted upon by an unbalanced force: • Explain Newton's First Law using the concept of inertia. • Use Newton's First Law to explain the motion of objects in a variety of contexts. • Describe and explain the motion of an object falling in a uniform gravitational field with air resistance. Newton's Second Law: If an unbalanced force acts upon an object, the object will accelerate in the direction of the net force. This can be given mathematically as: $\vec{a} = \frac{\vec{F}}{m}$ • Solve problems involving $\vec{F} = m\vec{a}$. • Explain the difference between mass and weight.	Refer to computer interactive 'Forces in 1 Dimension', https://phet.colorado.edu/. Discuss the concepts of weight and weightlessness in different contexts (e.g. on the surface of the Earth, on the surface of the Moon, inside a moving lift, in the International Space Station). Discuss the motion of spacecraft in an essentially frictionless environment. Study satellites in circular orbits with acceleration but constant speed. Discuss motion in different circumstances (e.g. on an inclined plane, in thick liquids, or for rigid objects). 
	Use computer Interactive 'Forces and Motion: Basics' and 'Forces and Motion', https://phet.colorado.edu/. Investigate Newton's First Law by moving objects of different mass over surfaces of very low friction (e.g. air tracks, ice, or layers of ball bearings). Investigate the relationships between terminal speeds and forces in a variety of contexts. Design and carry out individual or group investigations involving dropping objects in different fluids to observe and quantify non-uniform acceleration. 

Science Understanding	Possible contexts	
Newton's Third Law: When two objects interact, they exert forces on each other equal in magnitude and opposite in direction. The forces are identified in pairs, and the accelerations of each object will differ if the objects differ in mass. • Use Newton's Third Law to solve problems. • Identify pairs of forces in a variety of contexts, including the normal reaction force. • Describe and explain motion where Newton's Third Law occurs. • Use Newton's Laws to explain the motion of spacecraft. Undertake experiments to investigate the relationship between acceleration and either force or mass.	Use computer interactive 'Masses and Springs', https://phet.colorado.edu/. Investigate spring constants using Hooke's Law. Design experiments to investigate the effect of multiple springs. Students could test elastic bands, cooked spaghetti or noodles, or confectionery to compare differences in results. Use an air track, light gate, and slotted masses to investigate relationship between forces and acceleration. Investigate different types of air or water rockets. The design may be manipulated and the effect on the rockets' motion determined. Determine the coefficient of kinetic friction by measuring the acceleration of a moving object as it comes to rest. Design, build, and evaluate structures individually or in groups.	
	Investigate and assess the wide range of evidence from many sources that have contributed to the current understanding of motion. Investigate the influence that an understanding of the balance of forces has on the design of ancient and modern buildings, bridges, and other forms of engineering. Investigate and discuss the application of Newton's Laws of Motion in the development of various safety features involving people or objects in motion. Evaluate their social and economic impacts.	

Topic 2: Electric circuits

This topic extends students' knowledge and understanding of the concepts of circuit electricity. It explores the concept of electric charge and the requirements for electric current and introduces the concepts of potential difference, current, resistance, electric power, and efficiency. These concepts are applied to direct current (DC) electric circuits and form the essential understanding for Stage 2, Topic 2: Electricity and magnetism when discussing the production of magnetic fields and the generation and transmission of electricity.

Students extend their numeracy skills when problem solving in this topic, and their personal and social capability is fostered by considering electrical safety devices and the impact of electrical energy use on the local and global environment.

Subtopic 2.1: Potential difference and electric current

The existence of charged subatomic particles is used to explain the charging of objects. These ideas are extended to include potential difference and current, describing how the energy used to separate the charges enables an electric current in a closed circuit. The measurements of potential difference and current are introduced.

Science Understanding	Possible contexts	
Atoms contain positively charged protons and negatively charged electrons. Objects become charged when electrons are transferred from one object to another or redistributed on one object. Two like charges exert repulsive forces on each other, whereas two opposite charges exert attractive forces on each other. - Describe electric forces between like charges and between opposite charges. - Explain various phenomena involving interactions of charge. - Explain how electrical conductors allow charges to move freely through them, whereas insulators do not. Energy is required to separate positive and negative charges and this charge separation produces an electrical potential difference that can be used to drive current in circuits.	<i>This connects to the concept of work in Stage 1, Subtopic 4.1: Energy and Stage 2, Subtopic 2.2: Motion of charged particles in electric fields.</i> <i>It connects to the concepts of charge in Stage 2, Subtopic 2.1: Electric fields, and electric current used in Stage 2 subtopics 2.3: Magnetic fields and 2.5: Electromagnetic induction.</i>	
	Activities could include using: - a Van de Graaff generator - Perspex and ebonite rods - an electroscope - balloons. Investigate electrical discharges such as: - lightning - spark plugs - piezo igniters. Measure the total potential difference across a circuit and the potential difference across individual components.	

Science Understanding	Possible contexts	
The energy available to charges moving in an electrical circuit is measured using electric potential difference (voltage). This is defined as the change in potential energy per unit charge between two defined points in the circuit and is measured using a voltmeter. - Describe how a voltmeter is used in an electric circuit. - Explain the purpose of measuring potential difference in electric circuit. - Describe how electrical safety is increased through the use of: - fuses or circuit breakers - residual current devices.	Explore the application of potential differences. Examples include X-ray production in medical imaging and particle accelerators. Discuss the social and economic impacts, e.g. access to and affordability of such medical treatment in different parts of the world. Discuss innovative applications and limitations of semiconductors (e.g. photovoltaic cells in solar panels, LEDs) and superconductors (e.g. maglev trains, MRI). Explore beneficial and unexpected consequences of large-scale electricity production and transmission, taking into account that much of the world's energy production is used to provide the energy needed to drive electric current.	
	Research the safety aspects of working with electric circuits and the use of testing devices.	
Electric current is carried by discrete charge carriers. Charge is conserved at all points in an electrical circuit. Distinguish between electron current and conventional current. Electric current is the rate of flow of charge.	Use a water-flow analogy to facilitate students' understanding of this concept. Observe how the conductivity of metals, molten and aqueous ionic compounds, and ionised gases provides evidence for a variety of charge carriers.	
Solve problems involving $I = \frac{q}{t}$. An ammeter is used to measure the electric current at a point in a circuit. It is placed in series with the electrical component through which the current is to be measured.	Measure the total current in a circuit and the current at various places in the circuit.	

Subtopic 2.2: Resistance

Resistance is described and the factors affecting the resistance of a conductor are investigated. Ohm's Law is introduced, investigated, and applied. The concept of ohmic and non-ohmic conductors is also explored.

Science Understanding	Possible contexts	
Resistance for ohmic and non-ohmic components is defined as the ratio of potential difference across the component to the current in the component. The resistance of a conductor depends on its length, area of cross-section, temperature, and the type of the material of which it is composed. Resistance is constant for ohmic resistors, which conform to Ohm's Law. Ohm's Law states that current is directly proportional to the potential difference providing the temperature of the conductor remains constant. Solve problems involving $R = \frac{V}{I}$.	Computer interactive, 'Resistance in a Wire': http://phet.colorado.edu/en/simulation/resistance-in-a-wire. Ohm's Law is used when exploring electrical power, energy transmission, and the need for transformers.	
	Investigate the factors affecting the resistance of conductors such as the length of conductor, resistivity, or cross-sectional area. Investigate the relationship between current and potential difference for a variety of conductors, including graphical analysis.	
	Assess the economic, social, and environmental impacts of electrical safety devices, such as circuit breakers, fuses, and fuse wire.	

Subtopic 2.3: Circuit Analysis

Students are introduced to series, parallel, and composite (series and parallel) circuits and their construction. They analyse circuits to determine resistance, potential difference, and current.

Science Understanding	Possible contexts	
Circuit analysis and circuit design involve calculation of the potential difference across, the current in, and the power supplied to components in series, parallel, and composite circuits. The current is equal in each series component. - Solve problems involving $V_t = V_1 + V_2 + \dots V_n$ and $R_t = R_1 + R_2 + \dots R_n$ for components in series. The potential difference is equal across each parallel component. - Solve problems involving $I_t = I_1 + I_2 + \dots I_n$ and $\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \frac{1}{R_n}$ for components in parallel. Undertake experiments to investigate current, resistance, or potential difference in series and parallel circuits using various circuit elements.	Explore electric circuits using: http://www.physicsclassroom.com/Physics-Interactives/Electric-Circuits .	
	Students construct, test, and analyse a variety of electric circuits. Use circuit construction kits, e.g. from: http://phet.colorado.edu/en/simulation/circuit-construction-kit-ac.	

Subtopic 2.4: Electrical power

Students study electrical power, connecting Topic 2: Electric circuits to Subtopic 4.2: Momentum. Calculations of electric power enable connections to be made to Topic 3: Heat. The concept of efficiency is introduced in the context of electric circuits, enabling it to be applied in a range of situations in Subtopic 4.1: Energy. Concepts of power and efficiency are related to electricity production and transmission.

Science Understanding	Possible contexts	
Power is the rate at which energy is transformed by a circuit component. - Solve problems involving: $P = \frac{\Delta E}{t}$ $P = VI$ and the use of Ohm's Law formula. - Solve problems involving the cost of electrical energy, using kilowatt-hours. Electric circuits enable electrical energy to be transferred efficiently over large distances and transformed into a range of other useful forms of energy including thermal and kinetic energy, and light. - Solve problems involving: $\text{efficiency} = \frac{\text{useful energy}}{\text{total energy}} \times 100$ or $\text{efficiency} = \frac{\text{power output}}{\text{power input}} \times 100$	<i>This connects to the use of the concept power in Stage 1, Topic 4: Energy and momentum.</i> Explore electrical power and the use of appliances. Compare local and large-scale electricity generation in terms of their efficiency, convenience, and effect on the local and global environment.	
	Investigate the efficiency and running cost of a range of electrical appliances.	
	Analyse the energy losses that occur as electrical energy is fed through transmission lines from the generator to the consumer. Explore competing electric power transmission and storage systems such as large- and small-scale electricity production, and innovative home energy-storage technology. Evaluate benefits and limitations of each.	

Topic 3: Heat

In this topic, students extend their understanding of the concepts of energy, its transformations, transfer, and conservation by focusing on heat. Students explore the concepts of heat, temperature, thermal energy, and the different methods through which heat is transferred within a system. They study the change of state and the increase in temperature of a substance when heated from both qualitative and quantitative ways, extending their literacy and numeracy skills.

Possible contexts that can be used to develop these concepts include internal combustion engines, heating and cooling systems, and weather systems.

Students develop an understanding of the importance of heating and cooling systems for health and well-being, both personally and globally.

Subtopic 3.1: Heat and temperature

Students develop their understanding of the link between the temperature of matter and the kinetic energy of its particles. They investigate the flow of energy and explain it in terms of conduction, convection, and radiation. They explore applications of the expansion of materials due to the motion of their particles.

Science Understanding	Possible contexts	
Thermal energy is made up of the combined potential energy and the kinetic energy that is due to the vibration of the particles within the object. The particles within objects with higher temperatures have a higher average kinetic energy. An increase in the temperature of an object is due to an increase in its thermal energy. - Describe the links between temperature, vibrating particles, and thermal energy. Temperature can be measured with different scales (common ones being Celsius, Fahrenheit, and Kelvin). As the temperature decreases, the average kinetic energy of the particles drops until the lower limit (known as 'absolute zero') is reached. When a hotter object is put into contact with a cooler object, some of the thermal energy transfers from the hotter object to the cooler one. This <i>flow</i> of energy is referred to as 'heat'. If the objects remain in contact, then eventually the objects will reach the same temperature, putting the objects into 'thermal equilibrium'. - Describe heat as the flow of energy from hotter to cooler objects. - Describe thermal equilibrium.	Demonstrate and describe different temperature scales: Celsius, Fahrenheit, and Kelvin. Include those used to measure air temperature and the temperature of stars like the Sun. Discuss absolute zero. Compare the temperatures of a range of different objects, e.g. body temperature, boiling water, ice, light bulb, the Sun, liquid nitrogen, a warm room. Observe random motion of particles, using a smoke cell and light source and viewing through a microscope. Plot graphs of temperature vs time using thermometers made by students. Compare with commercially made thermometers. Use simulation experiments: https://phet.colorado.edu/en/simulation/en-ergy-forms-and-changes.	
	Conduct an investigation to determine the value of absolute zero. Compare the change in temperature in a volume of water when a small, hot object is added, as opposed to adding a large, warm object. Refer to http://youtu.be/wTi3Hn090Bs. Construct thermometers. These can be made with coloured liquids (alcohol) in thin tubes or straws in which the liquid expands as temperature increases. Investigate daily temperature changes using temperature maps on the Bureau of Meteorology website: www.bom.gov.au/jsp/awap/temp/index.jsp.	

Science Understanding	Possible contexts	
	Explore the international conventions involving different temperature scales and analyse the significance of using a consistent system. Examples may include Celsius, Fahrenheit, and Kelvin. Investigate the influence of engineering and technology on the development of the world's most sensitive thermometers	
Heat transfer can occur through conduction, convection, and radiation. • Explain how heat transfer can occur through conduction, convection, and radiation. • Describe examples of each heat-transfer process. Most solids, liquids, and gases expand when heated. • Describe applications of the expansion of matter due to heat transfer.	Use a conductivity star made of four different metals to demonstrate the different thermal conductivities of the metals. Show that water is a poor conductor by boiling water at the top of a test tube while ice held at the bottom of the test tube does not melt.	
	Explore the conduction of heat in a metal bar with small pieces of wax holding pins, and with one end of the bar being heated. (Graph distance and time.) Investigate convection currents with potassium permanganate crystals in water. Carry out investigations to explore gas and/or metallic expansion.	
	Analyse ways in which the use of poor conductors can have social and economic advantages (e.g. air in wool, synthetic, or down quilts; building insulation). Explore how an understanding of heat transfer during climate cycles (such as ocean currents or air currents) and their effects on weather can enable scientists to make predictions and design action for sustainability. <i>Note the link with Stage 2 Earth and Environmental Science, Topic 4: Climate change.</i>	

Science Understanding	Possible contexts	
	Explore the social, economic, and environmental impacts of applications of thermal expansion, such as: • thermostat control of heaters, irons, kettles • engineering of bridge building • turbines (particularly in power stations) driven by steam. Investigate changing technologies involving heat engines (e.g. steam engines (external combustion), diesel engines and petrol engines (internal combustion), Stirling engines) and innovations to reduce their environmental impact.	

Subtopic 3.2: Specific heat capacity

Students explore the amount of energy required to increase the temperature of various materials. This can then be linked to the properties and uses of materials.

Science Understanding	Possible contexts	
Energy can be added to or removed from a system without causing a change of state. The energy that is added or removed causes a change in temperature, ΔT. The change in temperature depends on the mass of the object, m, the amount of heat transferred to or from the object, Q, and the nature of the material (its 'specific heat capacity', c). These variables are linked through the formula: $Q = mc\Delta T$. - Describe and explain specific heat capacity. - Solve problems using the formula $Q = mc\Delta T$.	Experimental determination of specific heat capacity.	
	Investigate whether salt concentration affects the specific heat capacity of saltwater. Use calorimeters to investigate the specific heat capacity of common materials, such as cubes of wood, plastic, and metal.	
	Explore the significance to engineering of an understanding of specific heat capacity (e.g. for metal parts in an engine). Discuss the significance of the high specific heat capacity of water in climate homeostasis. <i>Note the link with Stage 2 Earth and Environmental Science, Topic 4: Climate change.</i>	

Subtopic 3.3: Change of state

Students explore the amount of energy absorbed or released during changes of state. They also explain changes of state using the particle model. The constant temperature during a change of state can be linked to applications.

Science Understanding	Possible contexts	
Matter commonly exists in three states: solid, liquid, and gas. To change a solid to a liquid (melting or fusion) and to change a liquid to a gas (boiling or vaporisation) requires the input of energy. This energy breaks bonds between atoms or molecules but does not change the temperature and is thus known as 'latent heat'. - Describe and explain latent heat. - Explain the difference between evaporation and boiling, using the particle model. The amount of latent heat required (Q) depends upon the nature of the substance (specifically, its latent heat capacity (L)) and the mass of the substance m, and is calculated using $Q = mL$. During the change of state from a gas to a liquid (condensation) or from a liquid to a solid (freezing or solidification), heat is released due to the formation of bonds between atoms or molecules. - Solve problems using the formula $Q = mL$. Some substances change from solid to gas (sublimation) or from gas to solid (deposition) without going through a liquid phase. Undertake experiments to determine the specific heat capacity or latent heat of different materials.	Discuss: - characteristics of the three common states of matter - change of state, using particle models and particle motion - other states of matter such as plasma. Discuss the difference between scalding burns and steam burns.	
	Investigate the latent heat of fusion of ice. Investigate the properties of liquids that make them suitable radiator coolants.	
	Explore how an understanding of body-temperature control mechanisms such as sweating can lead to innovations, for example, in sports science. Compare evaporative and refrigerative cooling of buildings and evaluate the benefits and limitations of each. Research pressure differences which can cause change of states or temperatures of refrigerants in refrigerators and evaluate its significance in food preservation. Explore the importance to life on Earth of water existing in all three states. <i>Note the link with Stage 1 Earth and Environmental Science, Topic 5: Importance of the hydrosphere.</i>	

Topic 4: Energy and momentum

This topic draws on content covered in Stage 1, Topic 1: Linear motion and forces and extends the study of motion to include energy and momentum. Conservation laws form the basis of many fundamental principles in physics, and a sound understanding of what these laws mean, and their implications, are essential to understand the physical world. This topic emphasises the law of the conservation of energy and the law of the conservation of momentum.

The concept of energy is used as a means to explain and predict the behaviours of different objects under different physical conditions. Students should be familiar with energy processes and transformations from previous learning in science. The first subtopic begins by exploring energy, work, and the relationship between the two. Students discuss abstract scientific definitions and how they might be used in the physical world. They consider different forms of energy, with a number of these forms suitable for quantitative analysis. Students also study the rate at which energy is used. As they discuss and explain these concepts, students further develop their literacy and numeracy skills.

Through the subtopic covering momentum, students extend their understanding of the relationship that exists between force and the motion of an object. There is a focus on interactions in one dimension between objects such as those that occur during collisions and explosions. This topic lends itself to a variety of experiments and investigations, including pendulum motion, and energy transfer and efficiency, enabling students to use their critical and creative thinking to deconstruct a problem, then formulate, review, and revise hypotheses.

Personal and social capability is strengthened by better understanding the physics of collisions and exploring innovative ways of increasing safety and reducing injury in vehicles and sports.

Subtopic 4.1: Energy

Students use mathematical relationships to determine and measure quantities based on work done, conservation of energy, and power. They investigate the efficiency of different mechanical systems and explore emerging technologies in sustainable energy generation.

Science Understanding	Possible contexts
The work done on an object is equivalent to the change in energy of that object. When a force is applied to an object causing a displacement over a distance, work is done. - Explain work in terms of an applied force. - Solve problems using $W = \Delta E$ and $W = Fs$ where the displacement is parallel to the force. Energy exists in a number of different forms. - Describe different forms of energy including kinetic, elastic, gravitational potential, rotational kinetic, heat, and electrical. Energy can be transferred from one object to another or transformed into different forms of energy. - Describe examples of energy being transferred from one object to another. - Describe examples of energy being transformed. - Explain qualitatively the meaning and some applications of various forms of energy, including kinetic energy and potential energy. - Solve problems using $E_K = \frac{1}{2}mv^2$ and $E_P = mgh$. - Describe energy transfers between objects and within different mechanical systems. Energy is conserved when transferred from one object to another in an isolated system. - Solve problems using the conservation of energy. - Describe and explain the energy losses that occur in systems involving energy transfers.	<i>This connects to the concept of energy used in Stage 1, Subtopic 6.4: Induced nuclear reactions and Stage 2, subtopics 2.2: Motion of charged particles in electric fields and 3.2: Wave-particle duality.</i> Investigate the work done on an object when the net force acting on the object is not in the direction of the displacement, using trigonometric calculations, $W = Fs \cos \theta$. Connect to work done when a projectile with air resistance moves through air. Show how the relationship $W = \Delta E$ can be derived using the definition of force and $E_K = \frac{1}{2}mv^2$. Demonstrate energy transfers using a steam engine, combustion engine, or other similar engines. Investigate the relationship between power and mechanical advantage using simple machines. <i>The concept of efficiency can be developed here; it is also covered in Stage 1, Subtopic 2.4: Electrical power.</i> Utilise computer interactive 'Energy Forms and Changes': https://phet.colorado.edu/. <i>The simple harmonic motion of pendulums links to Stage 1, Topic 5: Waves.</i> 
	Design and conduct individual or group experiments to determine the efficiency of different systems involving energy transfers. Test the conservation of energy by recording and measuring objects falling from different heights and recording their speed as they hit the ground. 

Science Understanding	Possible contexts	
Power is defined as the rate at which work is done and is equivalent to the rate at which energy is used. - Solve problems using $P = \frac{W}{t} \text{ and } P = Fv.$ - Interpret solutions in context.	Design and build a 'gravity car' – a car that uses a falling weight to transfer energy to wheel rotation. Investigate the motion of pendulums, demonstrating the law of conservation of energy.	
	Explore the difficulty of developing international conventions to define concepts such as energy and work to facilitate communication and collaboration. Assess the economic advantage of efficiency in different mechanical systems and any implications for developing or emerging technology such as regenerative brakes.	

Subtopic 4.2: Momentum

Students explore the relationship between force and momentum through this subtopic. They use the concept of momentum to reformulate the definition of a force as the rate of change of momentum, and examine impulse.

Students use the relationship between change in momentum (impulse) and force to explore the implications for existing technology, particularly in terms of safety in vehicles. They also apply this to spacecraft propulsion on Earth and the potential of propulsion in space, for example, for space probes and possible human space exploration. Students use the conservation of momentum to predict and explain physical phenomena, and may also use the conservation of momentum to investigate the forensic methods applied to analysing accidents. Students use technology to analyse and present experiments throughout this topic.

Science Understanding	Possible contexts
Momentum is a property of moving objects, which depends on their mass and velocity. Momentum can be expressed mathematically as $\vec{p} = m\vec{v}$. Momentum may be transferred from one object to another when a force acts over a time interval. The rate of change of momentum of an object with respect to time is equal to the net force acting upon the object. This can be expressed mathematically as: $\vec{F} = \frac{\Delta \vec{p}}{\Delta t}.$ The impulse of an object is equal to $F\Delta t$, and consequently equals the change in momentum. - Use Newton's Second Law in the form $\vec{F} = m\vec{a}$ to derive the formula: $\vec{F} = \frac{\Delta \vec{p}}{\Delta t}.$ - Solve problems involving changes in momentum and impulse (for one dimension). - Draw and interpret graphs of force vs time.	<i>This connects to the concept of momentum used in the Stage 2 subtopics 1.2: Forces and momentum and 3.2: Wave-particle duality.</i> Analyse the safety mechanisms of modern vehicles using concepts of impulse and momentum. Analyse the elasticity of bungee cords using the concept of impulse. <i>The focus in Stage 1 should be on one-dimensional situations. Two-dimensional situations are studied in Stage 2 Physics.</i> Compare the difficulty in stopping objects of different masses and speeds as a way to introduce the concept of momentum. Use simple experiments and demonstrations such as dropping eggs enclosed in student-designed cases to demonstrate basic concepts of impulse in context. Analyse the social and economic advantages of minimising the risk of injury by extending the time of a collision (e.g., by using seatbelts, air bags, bicycle helmets, boxing gloves, softball gloves, and safety nets). Explore applications in which an understanding of maximising the time during which the force acts leads to an increase in speed. Examples include: long rowing strokes, use of a woomera for throwing.

Science Understanding	Possible contexts	
	Evaluate ways in which increasing the gain in speed through maximising the force applied are used in sport science. Examples include the force applied to a ball using a tennis racquet, golf club, foot, or other object. Design a new piece of sport equipment.	
In an isolated system, the total momentum is conserved. - Use the conservation of momentum to solve problems in a variety of contexts. An elastic collision is one in which the total initial kinetic energy equals the total final kinetic energy. In an inelastic collision, some kinetic energy is transformed. - Describe the difference between an elastic collision and an inelastic collision using examples. - Solve problems involving one-dimensional collisions, using $E_K = \frac{1}{2}mv^2 \text{ and } \vec{p} = m\vec{v}.$ - Describe the energy transformations during inelastic collisions. Undertake experiments to investigate the conservation of energy or conservation of momentum.	<i>The focus in Stage 1 should be on one-dimensional situations. Two-dimensional situations are studied in Stage 2 Physics.</i> Use water-powered or air-powered rockets to describe the conservation of momentum. The activities may be extended to include systems of variable mass. Investigate Newton's cradle to explain that the conservation of momentum and the conservation of energy are both required to explain its motion. Refer to the computer interactive 'Collision Lab', https://phet.colorado.edu/.	
	Conduct experiments involving collisions, using various recording technology such as motion sensors and video. Use data and photographs from simulated vehicle accidents to make reasonable assumptions about their causes. Design and investigate collisions with minimal friction, using an air track with various recording technologies. Analyse video footage of explosions to further understand momentum in two dimensions. Investigate non-contact collisions by attaching magnets to trolleys.	
	Assess the technical and practical challenges associated with the development of spacecraft and the ways in which the conservation of momentum may be applied to spacecraft. Analyse the effect of the conservation of momentum in traffic accidents and hence determine factors that may have led to a collision. Evaluate the social and economic impacts.	

Topic 5: Waves

In this topic, students develop understanding of how the wave model can be used to describe, explain, and predict the transfer of energy through matter and space.

Students investigate a range of mechanical waves, and compare them with light waves. This leads to an understanding of a number of wave-related phenomena, including reflection, refraction, resonance, diffraction, polarisation, dispersion, and interference. Students also learn about the electromagnetic spectrum.

Possible contexts in which waves can be investigated include sound, music and acoustics, earthquakes and seismic waves; optics, lasers, and fibre optics; and wireless communications.

An insight into the rapidly increasing range of new technology for data storage and transmission extends students' personal and social capability as they recognise its global impact on health and well-being. They develop an understanding of the increased capacity of technology to communicate and explore some of the social, environmental, and economic impacts of scientific research in this area.

Subtopic 5.1: Wave model

Students develop their understanding of the wave model as a theory that can be used to describe the transfer of energy through matter and space. They investigate various properties of waves and the relationships between them.

Science Understanding	Possible contexts	
Waves are periodic oscillations that transfer energy from one point to another. In longitudinal waves, the direction of oscillation is parallel to the direction of travel of the wave. In transverse waves, the direction of oscillation is perpendicular to the direction of travel of the wave. - Represent transverse waves graphically and analyse the graphs. - Describe waves in terms of measurable quantities, including amplitude, wavelength (λ), frequency (f), period (T), and velocity (v). - Solve problems using: $f = 1/T$ $v = f\lambda$	<i>This connects to the concept of waves used in Stage 2 subtopics 3.1: Wave behaviour of light and 3.2: Wave-particle duality.</i> Show longitudinal and transverse waves in a slinky spring or wave machine. Discuss longitudinal waves in terms of compressions and rarefactions. Demonstrate and explain the Doppler effect and the formation of sonic booms at supersonic speeds. Use PhET, 'Sound and Waves', simulations: https://phet.colorado.edu/en/simulations/category/physics/sound-and-waves. Discuss the difference in time between seeing lightning and hearing thunder.	
	Investigate sound and vibrations (e.g. tuning fork in water, speaker cone). Analyse wave representations using technology, such as oscilloscopes or smartphone apps. Measure sound levels and relate them to the amplitude of the waves. Explore the relationships between the frequency and period of oscillators (such as pendulums and springs), and investigate factors that affect these quantities (such as string length). Investigate the velocity of various waves (sound, seismic, light) in various media (air, water, solids).	

Science Understanding	Possible contexts	
	Ascertain the economic, social, and environmental impacts of applications of the Doppler effect. Examples include: • detecting red-shift and blue-shift stars • radar guns • Doppler radar • reading weather patterns (e.g. to map a moving storm system using a stationary transmitter) • using sound waves to produce images of the heart (Doppler echocardiogram).	

Subtopic 5.2: Mechanical waves

Students use the wave model to describe and explain many phenomena that are observed in everyday life. They investigate wave behaviour and explore various applications of waves.

Science Understanding	Possible contexts	
Mechanical waves, such as sound and seismic waves, transfer energy through a physical medium. The natural frequency is the rate at which an object vibrates when it is disturbed by an outside force. A forced vibration occurs when a wave forces an object to vibrate at the same frequency as the wave. Resonance is the large-amplitude vibration that occurs in the object when the forced vibration is the same as its natural frequency. - Explain a range of wave-related phenomena, including echoes, refraction, and resonance, using the mechanical wave model. - Use the principle of superposition of waves to explain a range of interference phenomena, including standing waves and beats.	Test the frequency range of human hearing. Use an evacuated bell jar to show that sound does not travel through a vacuum. Use a large isolated solid wall to demonstrate echoes and reverberation. Discuss how bats use echolocation for navigation. Discuss the concept of harmonics in musical instruments. Demonstrate resonance with a singing rod. Investigate Tibetan singing bowls. Use the concept of resonance to tune a guitar. View videos of and discuss the Tacoma Narrows Bridge failure. Demonstrate the concept of beats, using tuning forks, musical instruments, or ICT. Visually compare the shape of the waves of different musical instruments (shown on an oscilloscope or other technology). Use PhET, 'Sound and Waves', simulations: https://phet.colorado.edu/en/simulations/category/physics/sound-and-waves.	
	Analyse standing waves mathematically using: $l = n \frac{\lambda}{2}$, for strings attached at both ends $\lambda = 2l/n$, where $n = 1, 2, 3, 4, \dots$ and l = length of the pipe for pipes open at both ends $\lambda = 4l/(2n-1)$, for pipes closed at one end.	

Science Understanding	Possible contexts	
	Investigate: - the effect of different tensions or lengths on the vibrating frequency, using, for example, strings or air columns - the effect of the density of the medium on the speed of sound - interference patterns in a ripple tank - the formation of standing waves in strings and pipes - the production of beats - the resonant frequency of an oscillator, such as a loudspeaker or spring, and the factors that affect the resonant frequency.	
	Evaluate the benefits, limitations, and ethical considerations of using technology in contexts such as: - voice recognition - medical imaging using ultrasound - sonar in submarines, in depth sounders, and for locating fish - acoustics and building design - resonance in built structures. Evaluate the significance of using seismic waves to determine the epicentre of an earthquake. <i>Note the link to Stage 1 Earth and Environmental Science, Topic 3: Processes in the geosphere.</i>	

Subtopic 5.3: Light

Students develop their understanding of the wavelike properties of light and other forms of electromagnetic radiation. They are introduced to the concept of oscillating electric and magnetic fields to model the behaviour of light and other forms of electromagnetic radiation. They investigate light and other forms of electromagnetic radiation and explore applications based on the wave model.

Science Understanding	Possible contexts
Light is the visible part of the electromagnetic spectrum — a spectrum that also includes radio waves, microwaves, infrared and ultraviolet radiations, X-rays, and gamma rays. Electromagnetic waves can be modelled as a transverse wave that can travel through a vacuum. Refraction is the change in direction of propagation of a wave as its speed changes. Diffraction is the bending/spreading of waves as they pass through an aperture or past a sharp edge. The plane of polarisation of an electromagnetic wave is the plane defined by the direction of travel and the oscillating electric field. - Describe reflection and refraction, using the ray model of light. - Explain a range of light-related phenomena, including reflection, refraction, total internal reflection, diffraction, and polarisation, using the wave model. Undertake experiments to investigate reflection or refraction of light using different media.	<i>This connects to the concept of waves used in Stage 2 subtopics 3.1: Wave behaviour of light and 3.2: Wave-particle duality.</i> Demonstrate transmission of a mobile phone signal into an evacuated bell jar. Observe spectra of various light sources through a spectrometer or prism. Refer to PhET, 'Light and Radiation', at: https://phet.colorado.edu/en/simulations/category/physics/light-and-radiation. Use Snell's Law to quantitatively describe refraction of light, using: $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2} = \frac{n_2}{n_1}$ Demonstrate transmission of light through optical fibres. Use a light box or laser to investigate reflection, refraction, total internal reflection, and the optics of concave and convex lenses and curved and flat mirrors. Investigate polarisation of light, using filters. Analyse the interaction between science and technology with advances in: - optics of camera lenses, telescopes, and binoculars - optometry (spectacles and corrective laser surgery) - the uses of optic fibres in medicine and communication - applications of polarisation (e.g. 3D glasses and sunglasses) - the uses of radio waves and microwaves in communication (e.g. Wi-Fi, mobile phones, and space communication)

Science Understanding	Possible contexts
	• heating using microwaves • the uses of X-rays and gamma rays in diagnostic and therapeutic medicine • laser airborne depth sounder (LADS). Assess the economic and health impacts of ultraviolet radiation exposure. 

Topic 6: Nuclear models and radioactivity

In this topic, students build on their understanding of the basic structure of the nucleus and the uses of radiation to develop an understanding of the concepts involved in the complex structure of the nucleus, stable and unstable nuclei, radioactivity, nuclear fission, and nuclear fusion. This understanding includes the concepts of nuclear force, nuclear reactions, radioactive decays, and mass–energy equivalence. They recognise that science is a global endeavour with significant contributions coming from many people.

Possible contexts that can be investigated include nuclear reactors, medical uses for radioisotopes, positron emission tomography, radioactive dating, high-energy particle accelerators, and developments in technology, engineering, and communication.

Students extend their ethical understanding by considering the impact of radioactive material on the environment and the importance of planning for the future, to protect and sustain the biosphere. They explore the way that science informs public debate about nuclear power and is in turn influenced by public debate.

Subtopic 6.1: The nucleus

Students develop an understanding of the structure of the atom and the nucleus, and the forces that exist within the nucleus. They also learn how to represent various nuclei.

Science Understanding	Possible contexts	
The basic structure of an atom comprises a small central nucleus consisting of protons and neutrons (nucleons) surrounded by electrons. Atomic nuclei can be described using their chemical symbol (X), mass number (A), atomic number (Z), and number of neutrons (N). A common representation is: ${}^A_Z\text{X}$. - Describe the structure of an atom, including the relative size and location of the nucleons and electrons. - Describe the structure of various nuclei from their symbol and vice versa. Isotopes are atoms of the same element that have different mass numbers. - Identify isotopes of an element based on their composition. - Explain why isotopes of the same element are chemically identical but have different physical properties. The nucleus is held together by a strong, attractive nuclear force. - Describe the properties of the strong nuclear force, including its short range. - Describe the balance between the electrostatic force and the strong nuclear force in stable nuclei. - Use the properties of the electrostatic force and the strong nuclear force to explain why some isotopes are unstable. - Locate stable and unstable nuclei on an N vs Z graph.	<i>This connects to the concept of the nucleus used in the Stage 2 subtopic 3.4: Standard Model.</i> Review the structure of the atom. Use online interactives to build an atom: https://phet.colorado.edu/en/simulation/build-an-atom. Explore other atomic models, the evidence that supported those models, and the limitations of each model.	
	Determine the value of the collaborative work of many scientists in the discovery of the nucleus and modifications made to their models in the light of new information. Research the benefits and limitations of the medical and industrial uses of different isotopes. Discuss monitoring and risk evaluation. Research how the ^{18}O content of water in ancient ice provides information about climate change and enables scientists to make predictions.	

Subtopic 6.2: Radioactive decay

Students learn about the factors that determine whether a nucleus is stable or unstable and undergoes radioactive decay. They also learn how the composition of the nucleus determines the type of decay that will occur.

Science Understanding	Possible contexts	
Unstable nuclei will undergo radioactive decay in which particles and/or electromagnetic radiation are emitted.	<i>This connects to the concept of the nucleus used in Stage 2 Subtopics 2.4: Motion of charged particles in magnetic fields and 3.4: Standard Model.</i> Discuss how large unstable nuclei may undergo nuclear fission rather than one of the three radioactive decays. Utilise the Australian Government ARPANSA website: http://www.arpansa.gov.au/radiationprotection/basics/radioactivity.cfm.	
	Explore examples of the collaborative work that has led to the discovery of radioactive elements and discuss the subsequent development of a range of applications.	
In alpha decay, an unstable nucleus emits an alpha particle, ${}^4_2\alpha$. Alpha decay typically occurs for nuclei with $Z > 83$. The general equation for an alpha decay is given by: ${}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2\alpha$. Write equations for the decay of heavy nuclei by alpha decay.	Simulate alpha decay using: https://phet.colorado.edu/en/simulation/alpha-decay. Demonstrate the use of a cloud chamber to detect alpha radiation. Investigate the decay series of unstable nuclei to form stable nuclei.	
	Explore the benefits and unexpected consequences of using alpha decay in, for example, quantum tunnelling (in a scanning tunnelling microscope) or in smoke alarms.	
In beta minus decay, an unstable nucleus emits an electron, ${}^0_{-1}e$. Beta minus decay occurs when a nucleus has an excess of neutrons, and involves the decay of a neutron into a proton, electron, and antineutrino. This is shown by the equation: ${}^1_0n \rightarrow {}^1_1p + {}^0_{-1}e + {}^0_0\bar{\nu}.$	Simulate beta decay using: https://phet.colorado.edu/en/simulation/beta-decay. Demonstrate the use of a cloud chamber to detect beta radiation.	
	Explore how the knowledge of conservation of momentum enables scientists to predict the presence of new particles such as neutrinos.	

Science Understanding	Possible contexts	
The general equation for beta minus decay of an unstable nucleus is shown by the equation: ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0e + {}_0^0\bar{\nu}_e.$ In beta plus decay, an unstable nucleus emits a positron, ${}_{+1}^0e$. Beta plus decay occurs when a nucleus has an excess of protons, and involves the decay of a proton into a neutron, positron, and neutrino. This is shown by the equation: ${}_Z^AX \rightarrow {}_{Z-1}^AY + {}_{+1}^0e + {}_0^0\nu_e.$ The general equation for beta plus decay of an unstable nucleus is given by: ${}_Z^AX \rightarrow {}_{Z-1}^AY + {}_{+1}^0e + {}_0^0\nu_e.$ - Describe the structure of unstable nuclei that causes each type of beta decay. - Write the equations for the decay of nuclei by beta minus and beta plus decay. - Use the conservation of charge to explain the emission of an electron in the decay of a neutron into a proton. - Use the conservation of charge to explain the emission of a positron in the decay of a proton into a neutron.	Evaluate the benefits and risks of the use of beta decay in industry and medicine.	
In gamma decay, an unstable nucleus emits high-energy gamma rays, γ. Gamma decay occurs when a nucleus is left with excess energy after an alpha or beta decay. The general equation for a gamma decay is given by: ${}_Z^AX^* \rightarrow {}_Z^AX + n\gamma$ where n is the number of high-energy gamma rays emitted. Write equations for the decay of unstable nuclei involving the emission of gamma rays.	Explore industrial, scientific, and medical uses of gamma emitters such as ${}^{99}\text{Tc}$ and assess the risks and their management.	

Science Understanding	Possible contexts	
The type of decay an unstable nucleus will undergo can be predicted based on the number of protons and neutrons within the nucleus. • Use the atomic and mass numbers to predict the type of decay for an unstable nucleus. • Use the location on an N vs Z graph to predict the type of decay for an unstable nucleus.		
The particles emitted in radioactive decay have sufficient energy to ionise atoms. The properties of the particles and/or radiation emitted in the different types of radioactive decay result in different penetration of matter. • Describe the effects of ionising radiation on living matter. • Describe methods of minimising exposure to ionising radiation. • Compare and contrast the ionising ability and penetration through matter of alpha, beta, and gamma radiations.	Discuss the protection for workers handling radioactive materials.	

Subtopic 6.3: Radioactive half-life

Students learn about the rate of radioactive decay. They explore how the rate of decay is related to the activity of radioactive samples, and the implications for managing radioactive materials. Students also investigate how the rate of decay can be used to determine the age of artefacts and other items.

Science Understanding	Possible contexts	
The number of radioactive nuclei in a sample of a given isotope decreases exponentially with time. Half-life is the time required for half of the radioactive nuclei in a sample to decay. The half-life of radioactive nuclei is independent of both the physical state and the chemical state of the material. The activity of a radioactive substance is the number of radioactive nuclei that decay per unit time. - Relate the activity of a sample to the number of radioactive nuclei present, and hence explain how it decreases exponentially with time. - Use data to estimate the half-life of radioactive nuclei. - Use data to estimate the activity or number of radioactive nuclei of a sample at different times. - Estimate the age of a sample based on the relative activity or the relative amounts of radioactive nuclei or their decay products. The range of products of nuclear decay, some with long half-lives, means that nuclear waste must be stored for long periods. - Explain the requirements for the safe storage of nuclear waste.	Investigate radioactive dating using: https://phet.colorado.edu/en/simulation/radioactive-dating-game. Refer to radiocarbon dating online interactive: http://www.pbs.org/wgbh/nova/tech/radiocarbon-dating.html.	
	Simulate radioactive decay using dice, or similar random events. Use a Geiger counter to detect radioactive emissions from different sources such as radioactive sources, bananas, concrete, bricks, and computers. Plot decay graphs and determine half-life.	
	Explore the benefits and limitations of radioactive dating, including: - carbon dating to determine the age of various artefacts and remains. - uranium-to-lead ratio to determine the age of rocks, comets. Research the issues associated with storing radioactive waste generated by the nuclear power and medical industries.	

Subtopic 6.4: Induced nuclear reactions

Students discuss the characteristics of induced nuclear fission reactions and apply them to the example of a nuclear fission reactor used for the generation of electrical power. The concept of mass–energy equivalence is used to explain the source of the energy produced in nuclear reactions.

Students also examine the fusion reactions in stars, and consider some advantages and disadvantages of fusion as a future source of power.

They explore nuclear reactions that are used to produce isotopes for scientific, medical, and industrial purposes. These isotopes can either be produced using neutrons from a nuclear reactor or using particle accelerators.

Science Understanding	Possible contexts
Nuclear fission can be induced in some heavy nuclei by the capture of a neutron. The nucleus splits into two nuclei and several neutrons. The total mass of the reactants in a fission reaction is greater than that of the products, releasing energy given by $E = \Delta mc^2$, where Δm is the mass of the reactants minus the mass of the products. - Calculate the energy released per fission reaction, given the relevant masses (in kg). On average, more than one neutron is emitted in nuclear fission. This leads to the possibility that these neutrons will induce further fissions, resulting in a chain reaction. - Relate the starting, normal operation, and stopping of a nuclear reactor to the nature of the chain reaction. The neutrons emitted as a result of nuclear fission have high speeds. ^{235}U undergoes fission with slow neutrons. Hence to induce fission in these nuclei the neutrons must be slowed down. Many neutrons are absorbed by surrounding nuclei, or escape and cause no further fissions. - Explain why neutrons have to be slowed down in order to produce fission in ^{235}U.	Demonstrate the chain reaction using a simulation, such as: www.phet.colorado.edu/en/simulation/nuclear-fission. Discuss how nuclear fission releases energy in the form of gamma rays and kinetic energy. Explain fission in terms of short-range nuclear-attractive forces and long-range electric-repulsive forces. Discuss how neutrons are slowed down as a result of collisions with a moderator. Explain why it is necessary to increase the fraction of ^{235}U in order to achieve a chain reaction because the fraction of ^{235}U in naturally occurring uranium is small. The process is called 'enrichment'. Assess the economic, social, and environmental impacts of the Manhattan Project. Investigate the way in which science informs public debate and is in turn influenced by public debate over the use of nuclear weapons. Analyse the economic and social benefits and the consequences of enrichment of isotopes of uranium from the ore. Explore the innovations which use neutron beams from nuclear reactors to produce radioisotopes for use in medicine and industry.

Science Understanding	Possible contexts	
Enrichment increases the proportion of ^{235}U in uranium fuel. - Describe how enrichment enables a chain reaction to proceed. - Use a diagram of a reactor to locate and discuss the function of the principal components of a water-moderated fission power reactor.		
Energy released during nuclear fission reactions can be harnessed for use in power generation. - Explain the use of nuclear fission in power production. - Describe some of the risks associated with the use of nuclear energy for power production.	Use a nuclear power plant simulator to gain an understanding of how the fission process is controlled. See: http://esa21.kennesaw.edu/activities/nukeenergy/nuke.htm.	
	Explore the beneficial or unexpected consequences of nuclear power and assess its monitoring, assessment, and risk. Explore public debate about nuclear power. Discuss some of the advantages and disadvantages of nuclear fission compared to fossil-fuel power stations.	
Nuclear fusion is the process in which two nuclei combine into a single nucleus. - Explain why high temperatures are needed for nuclear fusion to occur. The energy absorbed or released is given by $E = \Delta mc^2$, where Δm is the difference in mass between the reactants and the products. - Calculate the energy released per fusion reaction, given the relevant masses (in kg).	Compare the availability of fuel for fission and fusion reactions.	
	Examine the reasons it is not currently possible to use fusion for electricity generation. Compare the processes needed to safely control a fission reaction and a fusion reaction. Research attempts by scientists to produce sustainable fusion and predict possible outcomes. Research the birth and life cycle of stars and the detection of gravitational waves created by colliding black holes and assess the available evidence. Identify the conditions in the interiors of the Sun and other stars that allow nuclear fusion to take place, and hence how nuclear fusion is their main energy conversion process. Envisage the ways this may be harnessed. Discuss the advantages and disadvantages of nuclear fusion over nuclear fission as a future source of power.	

ASSESSMENT SCOPE AND REQUIREMENTS

Assessment at Stage 1 is school based.

EVIDENCE OF LEARNING

The following assessment types enable students to demonstrate their learning in Stage 1 Physics:

- Assessment Type 1: Investigations Folio
- Assessment Type 2: Skills and Applications Tasks.

For a 10-credit subject, students provide evidence of their learning through three or four assessments. Each assessment type should have a weighting of at least 20%.

Students complete:

- at least one practical investigation
- one investigation with a focus on science as a human endeavour
- at least one skills and applications task.

For a 20-credit subject, students provide evidence of their learning through six or seven assessments. Each assessment type should have a weighting of at least 20%.

Students complete:

- at least two practical investigations
- at least one investigation with a focus on science as a human endeavour
- at least two skills and applications tasks.

For both the 10-credit and 20-credit subjects, at least one assessment should involve collaborative work.

ASSESSMENT DESIGN CRITERIA

The assessment design criteria are based on the learning requirements and are used by teachers to:

- clarify for the student what they need to learn
- design opportunities for the student to provide evidence of their learning at the highest level of achievement.

The assessment design criteria are the specific features that:

- students should demonstrate in their learning
- teachers look for as evidence that students have met the learning requirements.

For this subject, the assessment design criteria are:

- investigation, analysis, and evaluation
- knowledge and application.

The specific features of these criteria are described below.

The set of assessments, as a whole, must give students opportunities to demonstrate each of the specific features by the completion of study of the subject.

Investigation, Analysis, and Evaluation

The specific features are as follows:

- IAE1 Deconstruction of a problem and design of a physics investigation.
- IAE2 Obtaining, recording, and representation of data, using appropriate conventions and formats.
- IAE3 Analysis and interpretation of data and other evidence to formulate and justify conclusions.
- IAE4 Evaluation of procedures and their effect on data.

Knowledge and Application

The specific features are as follows:

- KA1 Demonstration of knowledge and understanding of physics concepts.
- KA2 Application of physics concepts in new and familiar contexts.
- KA3 Exploration and understanding of the interaction between science and society.
- KA4 Communication of knowledge and understanding of physics concepts and information, using appropriate terms, conventions, and representations.

SCHOOL ASSESSMENT

Assessment Type 1: Investigations Folio

For a 10-credit subject, students undertake at least one practical investigation and one investigation with a focus on science as a human endeavour. Students may undertake more than one practical investigation within the maximum number of assessments allowed.

For a 20-credit subject, students undertake at least two practical investigations and at least one investigation with a focus on science as a human endeavour. Students may undertake more than two practical investigations and one science as a human endeavour investigation, within the maximum number of assessments allowed.

Students inquire into aspects of physics through practical discovery and data analysis, or by selecting, analysing, and interpreting information.

Practical Investigations

As students design and safely carry out investigations, they demonstrate their science inquiry skills by:

- deconstructing a problem to determine the most appropriate method for investigation
- formulating investigable questions and hypotheses
- selecting and using appropriate equipment, apparatus, and techniques
- identifying variables
- collecting, representing, analysing, and interpreting data
- evaluating procedures and considering their impact on results
- drawing conclusions
- communicating knowledge and understanding of concepts.

A practical investigation:

- enables students to investigate a question or hypothesis linked to one of the topics in Stage 1 Physics
- may provide the opportunity for a range of both individual and collaborative activities.

At least one practical investigation should enable students to individually deconstruct a problem, for which the outcome is uncertain, and design their own method and justify their plan of action.

Evidence should outline the deconstruction process, the method designed as most appropriate, and a justification of the plan of action. This should be presented on a maximum of four sides of an A4 page (minimum font size 10). Suggested formats for this evidence include flow charts, concept maps, tables, or notes. This evidence must be submitted with the practical report.

In order to manage the implementation of an investigation efficiently, students could individually design investigations and then conduct one of these as a group, or design hypothetical investigations at the end of a practical activity.

For practical investigations, students present an individual report.

A practical report must include:

- introduction with relevant physics concepts, and either a hypothesis and variables, or an investigable question
- materials/apparatus
- the method that was implemented
- identification and management of safety and/or ethical risks
- results, including table(s) and/or graph(s)
- analysis of results, including identifying trends and linking results to concepts
- evaluation of procedures and their effect on data, and identifying sources of uncertainty
- conclusion, with justification.

The report should be a maximum of 1000 words if written, or a maximum of 6 minutes for an oral presentation, or the equivalent in multimodal form.

Only the following sections of the report are included in the word count:

- introduction
- analysis of results

-
- evaluation of procedures
 - conclusion and justification.

Suggested formats for presentation of a practical investigation report include:

- a written report
- an oral presentation
- a multimodal product.

Science as a Human Endeavour Investigation

Students investigate a contemporary example of how science interacts with society. This may focus on one or more of the key concepts of science as a human endeavour described on pages 11 and 12 and may draw on a context suggested in the topics or relate to a new context.

Students could consider, for example, how:

- humans seek to improve their understanding and explanation of the natural world
- working scientifically is a way of obtaining knowledge that allows for testing scientific claims
- scientific theory can change in the light of new evidence
- technological advances change ways of working scientifically
- links between advances in science impact and influence society
- society influences scientific research
- physics is important in sport
- emerging physics-related careers and pathways involve science
- 'blue sky' research leads to new technologies.

Students access information from different sources, select relevant information, analyse their findings, and explain the connection to science as a human endeavour.

Possible starting points for the investigation could include, for example:

- the announcement of a discovery in the field of physics
- an expert's point of view on a controversial innovation
- a TED talk based on a development in physics
- an article from a scientific publication (e.g. *Cosmos*)
- public concern about an issue that has environmental, social, economic, or political implications.

Based on their investigation, students prepare a scientific communication, which must include the use of scientific terminology.

The communication should be a maximum of 1000 words if written, or a maximum of 6 minutes for an oral presentation, or the equivalent in multimodal form.

For this assessment type, students provide evidence of their learning in relation to the following assessment design criteria:

- investigation, analysis, and evaluation
- knowledge and application.

Assessment Type 2: Skills and Applications Tasks

For a 10-credit subject, students undertake at least one skills and applications task. Students may undertake more than one skills and applications task within the maximum number of assessments allowed, but at least one should be under the direct supervision of the teacher. The supervised setting (e.g. classroom, laboratory, or field) should be appropriate to the task.

For a 20-credit subject, students undertake at least two skills and applications tasks. Students may undertake more than two skills and applications tasks within the maximum number of assessments allowed, but at least one should be under the direct supervision of the teacher. The supervised setting (e.g. classroom, laboratory, or field) should be appropriate to the task.

Skills and applications tasks allow students to provide evidence of their learning in tasks that may:

- be applied, analytical, and/or interpretative
- pose problems in new and familiar contexts
- involve individual or collaborative assessments, depending on task design.

A skills and applications task may involve, for example:

- solving problems
- designing an investigation to test a hypothesis or investigable question
- considering different scenarios in which to apply knowledge and understanding
- graphing, tabulating, and/or analysing data
- evaluating procedures and identifying their limitations
- formulating and justifying conclusions
- representing information diagrammatically or graphically
- using physics terms, conventions, and notations.

As a set, skills and applications tasks should be designed to enable students to apply their science inquiry skills, demonstrate knowledge and understanding of key physics concepts and learning, and explain connections with science as a human endeavour. Problems and scenarios should be set in a relevant context, which may be practical, social, or environmental.

Skills and applications tasks may include, for example:

- modelling or representing concepts
- developing simulations
- practical and/or graphical skills
- a multimodal product
- an oral presentation
- participation in a debate
- an extended response
- responses to short-answer questions
- a structured interview
- an excursion report
- a response to science in the media.

For this assessment type, students provide evidence of their learning in relation to the following assessment design criteria:

- investigation, analysis, and evaluation
- knowledge and application.

PERFORMANCE STANDARDS

The performance standards describe five levels of achievement, A to E.

Each level of achievement describes the knowledge, skills and understanding that teachers refer to in deciding how well students have demonstrated their learning on the basis of the evidence provided.

During the teaching and learning program the teacher gives students feedback on their learning, with reference to the performance standards.

At the student's completion of study of a subject, the teacher makes a decision about the quality of the student's learning by:

- referring to the performance standards
- taking into account the weighting of each assessment type
- assigning a subject grade between A and E.

Performance Standards for Stage 1 Physics

Investigation, Analysis and Evaluation		Knowledge and Application
A	Critically deconstructs a problem and designs a logical and coherent physics investigation with detailed justification. Obtains, records, and represents data, using appropriate conventions and formats accurately and highly effectively. Systematically analyses and interprets data and evidence to formulate logical conclusions with detailed justification. Critically and logically evaluates procedures and their effect on data.	Demonstrates deep and broad knowledge and understanding of a range of physics concepts. Applies physics concepts highly effectively in new and familiar contexts. Critically explores and understands in depth the interaction between science and society. Communicates knowledge and understanding of physics coherently, with highly effective use of appropriate terms, conventions, and representations.
B	Logically deconstructs a problem and designs a well-considered and clear physics investigation with reasonable justification. Obtains, records, and represents data, using appropriate conventions and formats mostly accurately and effectively. Logically analyses and interprets data and evidence to formulate suitable conclusions with reasonable justification. Logically evaluates procedures and their effect on data.	Demonstrates some depth and breadth of knowledge and understanding of a range of physics concepts. Applies physics concepts mostly effectively in new and familiar contexts. Logically explores and understands in some depth the interaction between science and society. Communicates knowledge and understanding of physics mostly coherently, with effective use of appropriate terms, conventions, and representations.
C	Deconstructs a problem and designs a considered and generally clear physics investigation with some justification. Obtains, records, and represents data, using generally appropriate conventions and formats, with some errors but generally accurately and effectively. Undertakes some analysis and interpretation of data and evidence to formulate generally appropriate conclusions with some justification. Evaluates procedures and some of their effect on data.	Demonstrates knowledge and understanding of a general range of physics concepts. Applies physics concepts generally effectively in new or familiar contexts. Explores and understands aspects of the interaction between science and society. Communicates knowledge and understanding of physics generally effectively, using some appropriate terms, conventions, and representations.
D	Prepares a basic deconstruction of a problem and an outline of a physics investigation. Obtains, records, and represents data, using conventions and formats inconsistently, with occasional accuracy and effectiveness. Describes data and undertakes some basic interpretation to formulate a basic conclusion. Attempts to evaluate procedures or suggest an effect on data.	Demonstrates some basic knowledge and partial understanding of physics concepts. Applies some physics concepts in familiar contexts. Partially explores and recognises aspects of the interaction between science and society. Communicates basic physics information, using some appropriate terms, conventions, and/or representations.

Investigation, Analysis and Evaluation		Knowledge and Application
E	Attempts a simple deconstruction of a problem and a procedure for a physics investigation. Attempts to record and represent some data, with limited accuracy or effectiveness. Attempts to describe results and/or interpret data to formulate a basic conclusion. Acknowledges that procedures affect data.	Demonstrates limited recognition and awareness of physics concepts. Attempts to apply physics concepts in familiar contexts. Attempts to explore and identify an aspect of the interaction between science and society. Attempts to communicate information about physics.

ASSESSMENT INTEGRITY

The SACE Assuring Assessment Integrity Policy outlines the principles and processes that teachers and assessors follow to assure the integrity of student assessments. This policy is available on the SACE website (www.sace.sa.edu.au) as part of the SACE Policy Framework.

The SACE Board uses a range of quality assurance processes so that the grades awarded for student achievement in the school assessment are applied consistently and fairly against the performance standards for a subject, and are comparable across all schools.

Information and guidelines on quality assurance in assessment at Stage 1 are available on the SACE website (www.sace.sa.edu.au).

SUPPORT MATERIALS

SUBJECT-SPECIFIC ADVICE

Online support materials are provided for each subject and updated regularly on the SACE website (www.sace.sa.edu.au). Examples of support materials are sample learning and assessment plans, annotated assessment tasks, annotated student responses, and recommended resource materials.

ADVICE ON ETHICAL STUDY AND RESEARCH

Advice for students and teachers on ethical study and research practices is available in the guidelines on the ethical conduct of research in the SACE on the SACE website (www.sace.sa.edu.au).